

Reconstruction of objects from grasps with latent diffusion modeling

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Abstract—The goal of this project is to design and train a latent diffusion model that takes geometric information from a anthropomorphic robotics hand grasping an object as an conditional input and reconstructs said object based on the information provided by the input. Research questions of this project are, how much geometric information is encoded in the grasp of a robotic hand and how can this information be used to reconstruct the grasped object. Which latent space encoding and variational autoencoder architecture achieves the best compromise between preserving the information of the original input data and accelerating the training of the model. Which diffusion model architecture offers the best reconstruction quality. Which conditioning technique is best suited for guiding the diffusion process to generate object shapes based on the grasp input. How should the training data be preprocessed or enhanced to improve the training performance of the latent diffusion model.

Keywords—*Latent Space, Diffusion, 3d-object reconstruction, Generative modeling, Humanoid robotics*

I. INTRODUCTION

Grasping objects is one of the most important tasks in robotics. Recent data-driven approaches tried to solve this task by synthesizing grasps with generative artificial intelligence (AI) like variational autoencoders (VAE) or diffusion models from the geometry of the object as an input [1–3]. While being effective in generating a diverse set of possible grasps, these models also generate many invalid grasps and often rely on optimization techniques or discriminator networks to detect invalid grasps. This problem arises from the incomplete geometric representation of the input, because only the front of the object is seen by the robot. To improve the success rate of valid grasp generation recent works investigated neural shape completion techniques from partial geometric input [4, 5]. These works relied on vision-based systems to provide the partial geometric information and showed a reconstruction quality. But for industrial applications robots often have to be cheap and are often not equipped with camera systems. Furthermore, robots in an industrial setting often have to work in darkness or in small occluded work spaces where vision cannot be provided. To resolve this issue other works explored the idea of reconstructing object shapes from tactile sensing [6, 7]. These works rely tactile sensors that generate a pressure image of the contact surface and extract features with vision transformers or convolutional networks from the images with great effectiveness. But we are interested in the task of shape completion based on internal geometric information like the joint angles and not any geometric information provided by sensors as there has been done little research in this direction. So our project asks the question how much geometric information is encoded in the interaction between the robot hand and the object and is it possible to reconstruct valid object shapes from that information. The closest work that investigated this idea is UGG from [8]. In UGG the authors

introduced a unified latent space where object O and grasp G where modeled as a joint probability $P(G, O)$. The authors show that this latent space allows for training both the conditional generation of objects and grasps. While this approach is interesting, we want to develop a direct strategy by directly training the diffusion model to learn the conditional probability of generating objects from a given grasp $P(O|G)$ and compare our results to UGG.

II. TECHNICAL OUTLINE

Our main idea is inspired by works like [9, 10] who showed that the DDPM [11] process can be performed in the latent space to accelerate the training, while retaining very high state of the art generation quality. As we are dealing with high-dimensional 3d Data and have limited computing resources latent diffusion will be key in giving us good generation quality for a reasonable amount of computing cost. Other works have proven that latent diffusion modeling for 3d point cloud data achieves good results [8, 12, 13]. So we want to stick with point clouds as well, as there is much groundwork that we can rely on from past works and point clouds represent a budget friendly way of modeling the surface of a 3d object. For latent diffusion the VAE is of great importance as it generates the latent codes to perform the diffusion on. For our work we would like to experiment with different VAEs. Pointnet++ and PVCNN [14, 15] are popular and easy to use autoencoders that generate global shape latents. LION [12] a more advanced architecture that suits our needs perfectly constructs both a global shape latent and a point structure latent that better retains the geometrical structure of the original point cloud, thus providing the diffusion model with more geometric information. Then the diffusion model itself must be considered. Traditional latent diffusion models leverage a U-Net Design that relies on convolution layers [9, 10], but there was also much work done on diffusion transformers for point cloud generation recently [7, 13, 16]. These works showed that transformers are very good in preserving the shape of the original object as the point cloud can be tokenized and enhanced with a positional encoding and fed to the transformer. Enhancing the input and feeding the model different input modalities like the joint angles, a point cloud of the robot hand or contact points between hand and object, will also be investigated. But for the start of the project, we will keep the model as simple as possible to have a solid baseline that can be improved upon. For the first two weeks we will explore the provided training data and try out VAEs with pretrained weights. Then we will develop a simple architecture baseline of VAE that generates a global latent. Perform diffusion on the latent and let the model learn to denoise the latent. Then feed the denoised latent to the decoder. So, the first baseline will be unconditional generation of shapes and two ease the problem

we could stick to only one object category at first, thus reducing the problem to a unimodal distribution, and try to overfit the model to a small dataset. If the model training loss improves and the model shows that it can overfit to a small easy dataset, we can iteratively increase the dataset size until we can be sure that the model can learn the distribution. If for instance, the model shows no convergence, then we can increase the model capacity or switch the architecture. If the model shows good convergence during training and

satisfying reconstruction quality we can add more features, for example grasp-conditional object reconstruction. We can try out different architectures with this approach compare them and when we finalized the architecture, we can experiment with more advanced techniques like cross-attention as a conditioning mechanism. Feeding the network different input modalities. Trying out implicit diffusion [17] an so on.

III. REFERENCES

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